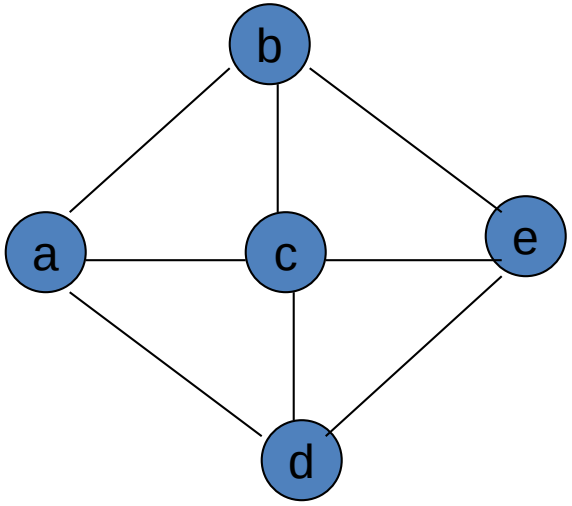
Homework: CSP1.

Consider the constraint graph on the right.
The domain for every variable is [1,2,3,4], i.e., each node is assigned one of 1,2,3,4.
There are 2 unary constraints:
- variable “a” cannot take values 3 and 4.
- variable “b” cannot take value 4.

There are 8 binary constraints stating that variables connected by an edge cannot have the same value.

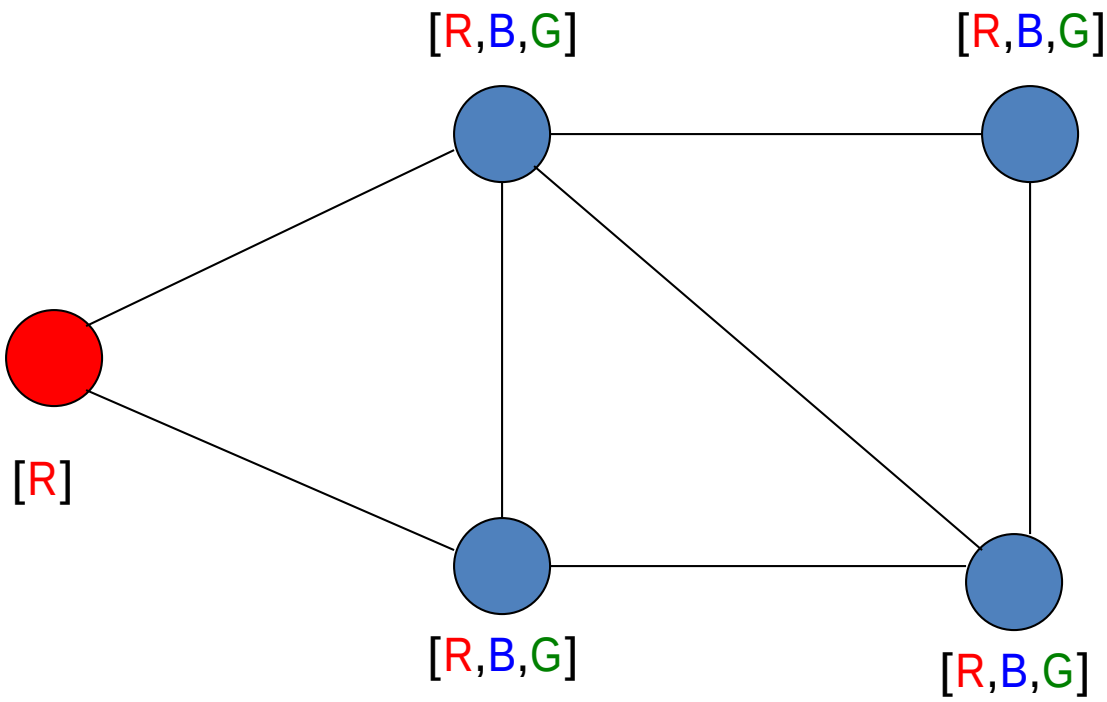
Find a solution for this CSP by using the following heuristics: minimum value heuristic, degree heuristic, forward checking. Explain each step of your answer.

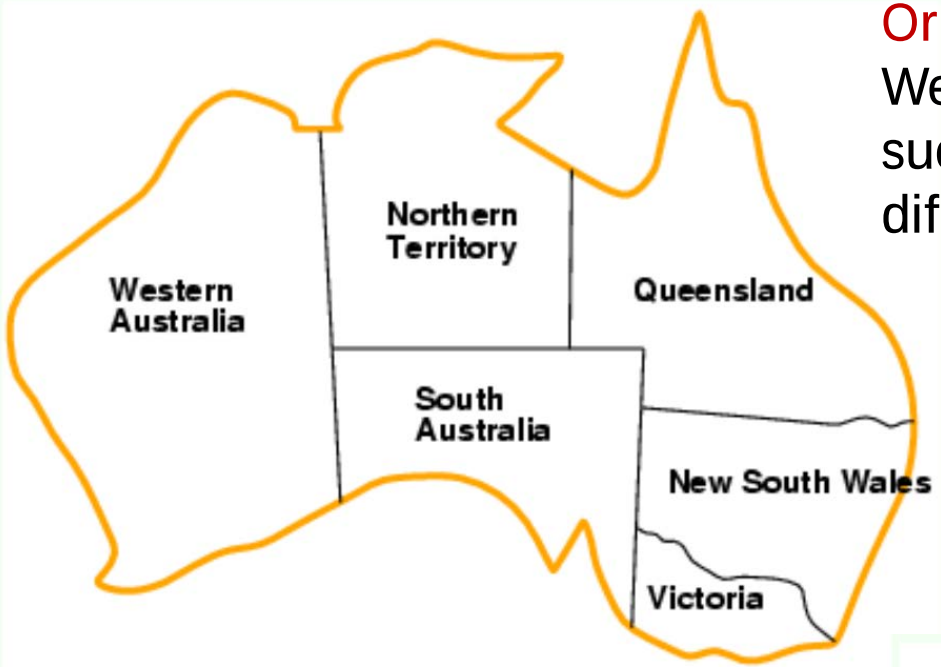


CONSTRAINT GRAPH

Homework: CSP2

Can you solve this coloring problem?





Original Australia map.

We can use 3 colors to color the map such that neighboring states have different colors.

Homework: CSP3

Suppose Australia map is changed to this. Show that we need 4 colors to color this map.

